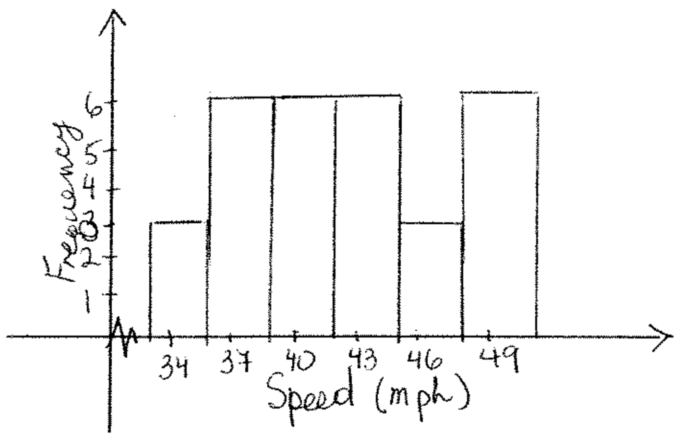
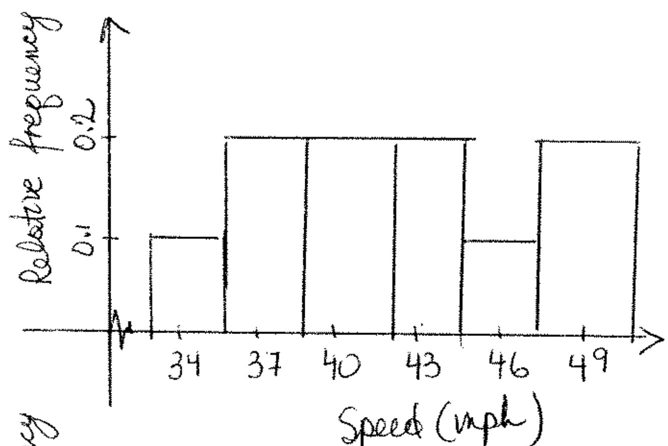
SOLUTIONS (Chapter 2)

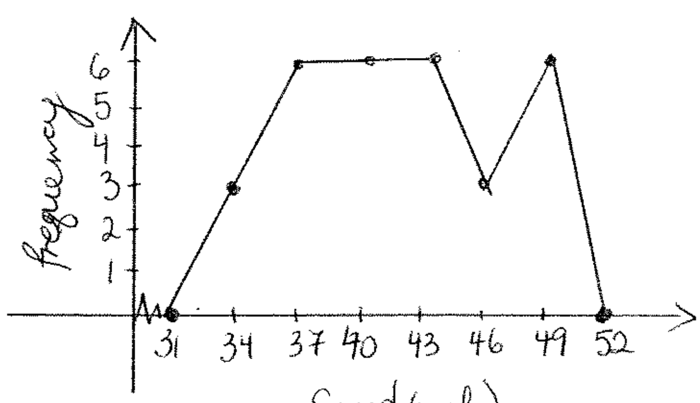
1)
min = 33
max = 50

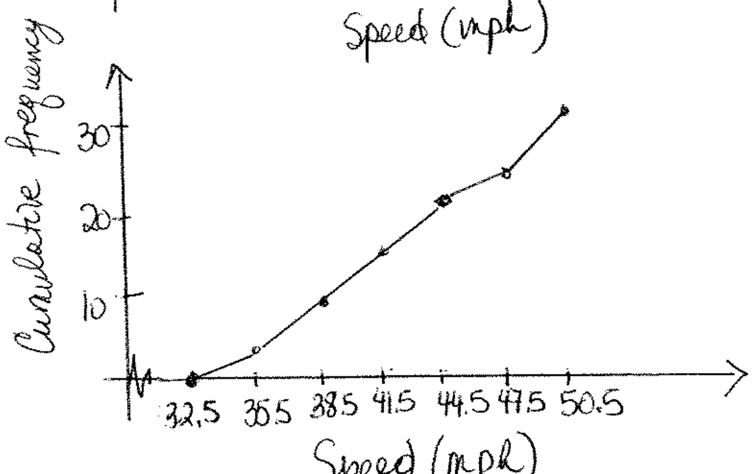
$$\frac{50-33}{6} = \frac{17}{6} = 2.5 \rightarrow 3$$

CLASS Speed (mph)	Frequency	Relative f	Cumulative f	Midpoint
33-35	3	3/30 = 0.1	3	34
36-38	6	6/30 = 0.2	9	37
39-41	6	0.2	15	40
42-44	6	0.2	21	43
45-47	3	0.1	24	46
48-50	6	0.2	30	49







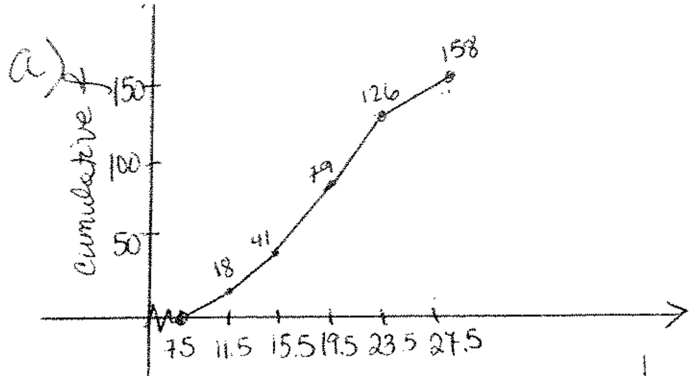


over 41 mph → add relative frequencies for classes: 42-44, 45-47, 48-50

$0.2 + 0.1 + 0.2 = 0.5 = \boxed{50\%}$

2)

a)



b) use midpoints. (9.5, 13.5, 17.5, 21.5, 25

$$\bar{x} = \frac{9.5\times 18 + 13.5\times 23 + 17.5\times 38 + 21.5\times 47 + 25.5\times 32}{158}$$

$$\bar{x} = \frac{2973}{158} = \underline{18.81}$$

3) min = 12

max = 26

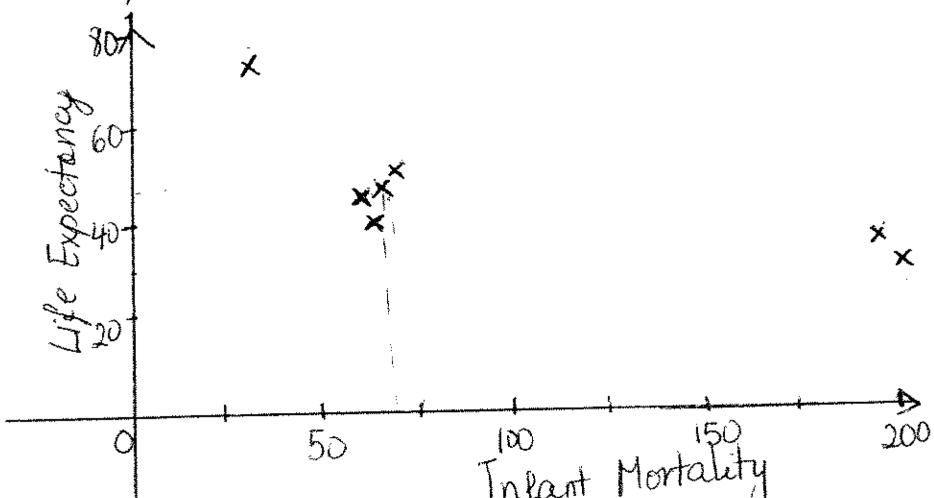
$\frac{26-12}{8} = 1.75 \to 2$ class width

Class	Tally	Frequency	Cumulative Freq.	Relative freq.
12-13	||||	4	4	4/40 = 0.1
14-15	||||	4	8	0.1
16-17	|||	3	11	0.075
18-19	~~||||~~ ~~||||~~ |	11	22	0.275
20-21	~~||||~~	5	27	0.125
22-23	||||	4	31	0.1
24-25	~~||||~~ ||	7	38	0.175
26-27	||	2	40	0.05

top 90% → all but class 12-13; minimum score = 14

29/40 = 0.725 = 72.5%

4)



As the number of infant mortality decreases, the life expectancy increases

OR

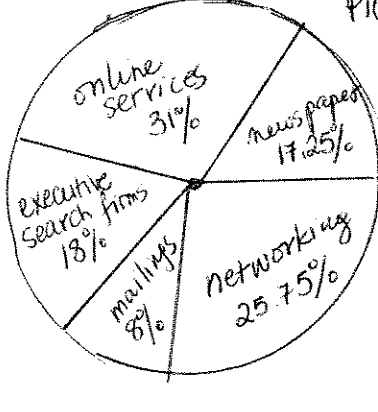
as infant mortality ↑ the life expectancy ↘

5) Find relative frequencies/angles; then:

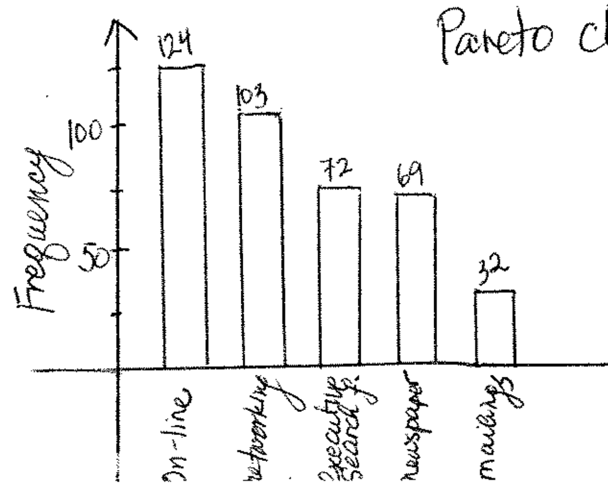
in order

%	∡
7.25	62.1°
31	111.6°
8	64.8°
8	28.8°
25.75	92.7°

Pie chart



Pareto chart



(6) a) 80
b) 80-35=45

(7) 3 | 8 6 6 5 7 9 8 6 5 3

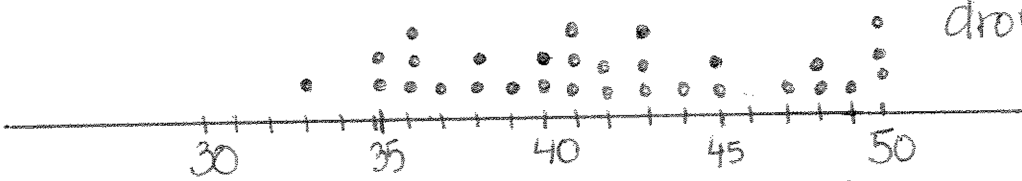
4 | 4 1 3 2 9 8 0 1 3 5 5 1
7 0 2 3 8
5 | 0 0 0

more then 50% of the cars drove 40-49 mph.

You can order the data & have 2 stems.

3 | 3
3 | 5 5 6 6 6 7 8 8 9
4 | 0 0 1 1 1 2 2 3 3 3 4
4 | 5 5 7 8 8 9
5 | 0 0 0
5 |

11/30 = 36.7% of the cars drove 40-44 mph



mean = $\frac{1255}{30} = 41.8\overline{3}$
$= 41.8$ mph

median = $\frac{44+42}{2} = 41.5$ mph

mode - does not have much meaning since there are 4 values with $f=3$

(8) - to approximate the mean using 5 classes → use midpoints
approximation: mean = 124.3
- mean = 124.2
median = 124
mode = 125
98 is an outlier

(9) 2.75

(10) 83.6

(11) 70.6

(12) 31

(13) a) $\bar{x} = \frac{227}{9} = 25.2$
$s = 17.8$

b) $\bar{x} = 70.3$
$s = 1.49$

(14) range = 4.4, $\bar{x} = 3.12$
$s^2 = 3.3124$
$s = 1.82$

(15) lower standard deviation → data more consistent
Choose Type A

(16) 95%

(17) 2.5%

(18) a) At least 75% of the heights should fall between 58.6 in & 68.6 in.
b) NO
c) At least 88.9% between 56.1 & 71.1 in.

(19) $s = 5.1$